

EENG250 ELECTRIC CIRCUITS I · CHAPTER 2

Circuit Elements

Sources, resistors and the three laws that solve every circuit in this course:
Ohm's law, KCL and KVL.

Prepared by Dr. Adel Daaboul

PREDICT FIRST

Two 1100 W heaters on 220 V. Series or parallel: which gives more heat, and by how much?

TODAY

By the end of this chapter you can

1 · Sources

Use ideal, independent and dependent sources, and say which connections are allowed.

2 · Ohm's law

Write $v = iR$ with the right sign, and find the power in a resistor.

3 · Kirchhoff's laws

Write KCL at a node and KVL around a closed path.

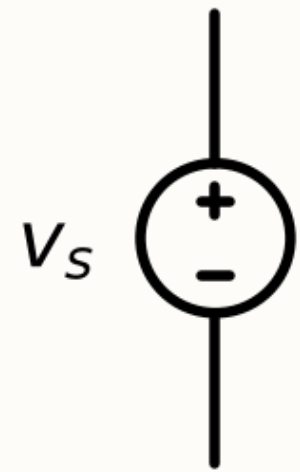
$$\sum i = 0, \quad \sum v = 0$$

4 · Analysis

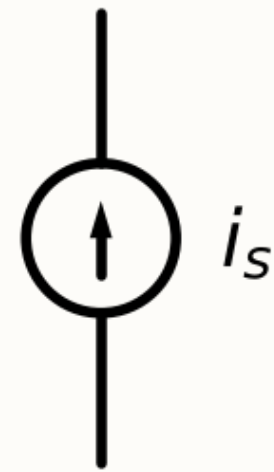
Solve circuits with dependent sources and check them with a power balance.

BASIC ELEMENTS

The first three of the five ideal elements



ideal voltage
source



ideal current
source



resistor

Sources and resistors are enough for Chapters 2 to 5. The other two ideal elements, inductors and capacitors, come in Chapter 6. An ideal element is a model, not a part you can buy.

IDEAL SOURCES

An ideal source holds one quantity fixed, whatever the other does

Ideal voltage source

Keeps its voltage whatever current flows. Its current is set by the rest of the circuit.

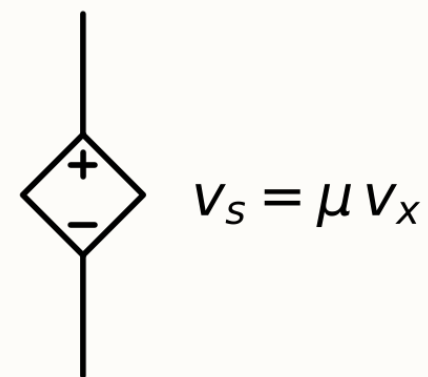
Ideal current source

Keeps its current whatever voltage appears across it. Its voltage is set by the rest of the circuit.

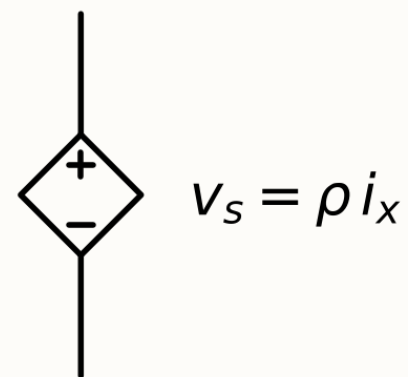
Independent: the value is fixed. **Dependent:** the value is set by a voltage or current somewhere else in the circuit.

DEPENDENT SOURCES

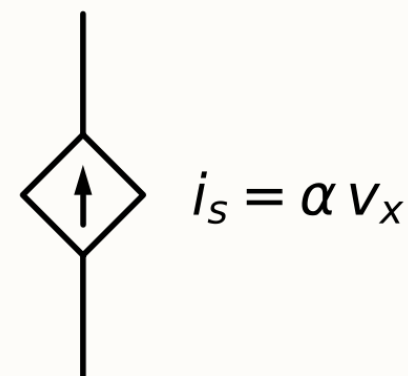
A diamond marks a dependent source



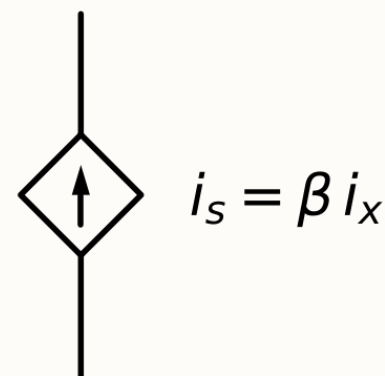
(a) voltage-controlled voltage source



(b) current-controlled voltage source



(c) voltage-controlled current source

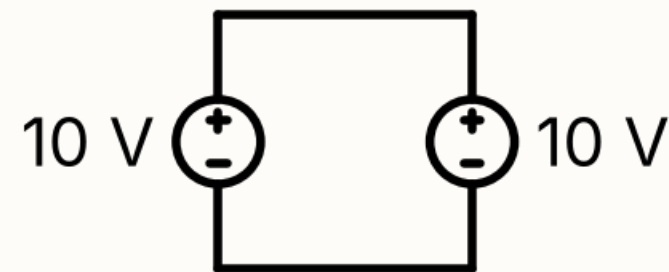


(d) current-controlled current source

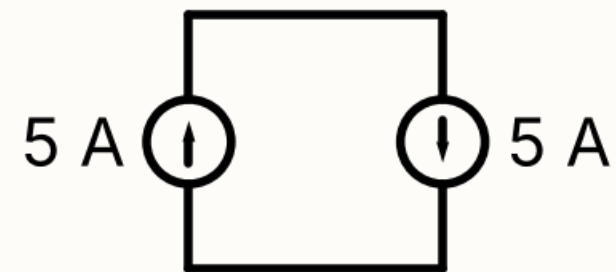
Gain	Unit
μ (a)	none
ρ (b)	$V/A = \Omega$
α (c)	$A/V = S$
β (d)	none

The diamond tells you it is dependent; the + and – or the arrow tells you whether it sets a voltage or a current.

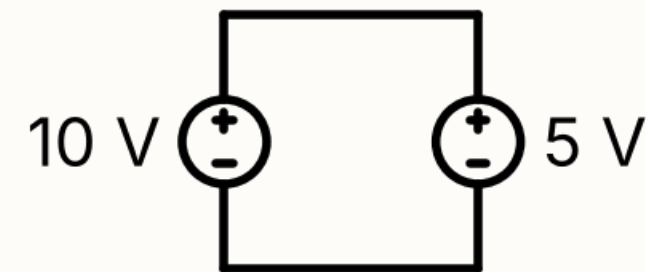
Which of these connections are not allowed?



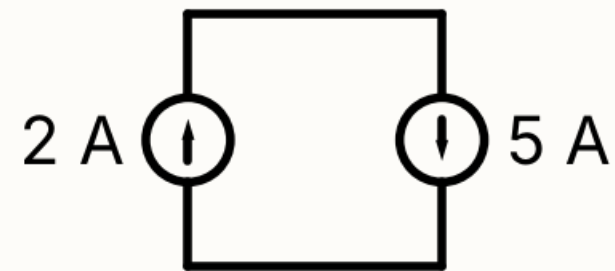
(a)



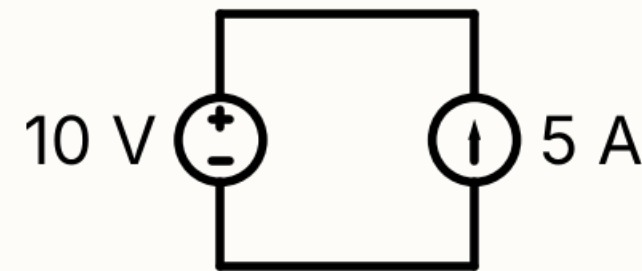
(b)



(c)



(d)



(e)

Ask of each source: what does it insist on?

Can both sources get their way at the same time?

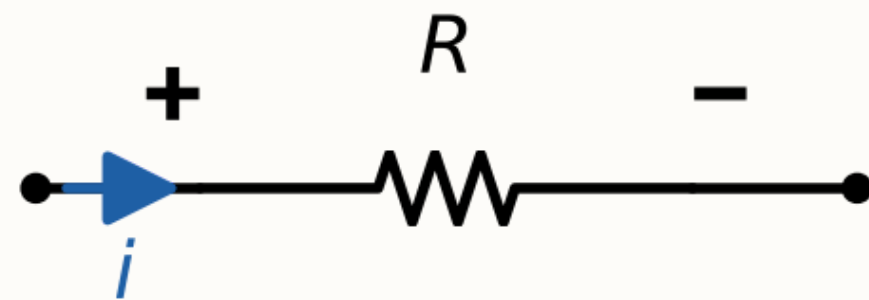
ANSWER

(c) and (d) are not allowed

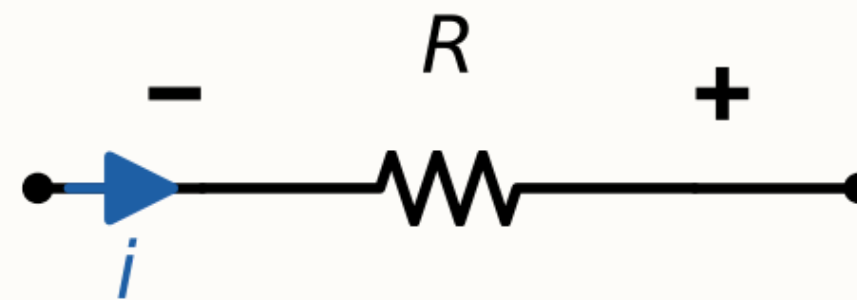
Connection	Allowed?	Why
(a) 10 V with 10 V	Yes	Both insist on 10 V, same polarity
(b) 5 A with 5 A	Yes	Both push 5 A the same way round the loop
(c) 10 V with 5 V	No	One pair of terminals cannot be at 10 V and 5 V
(d) 2 A with 5 A	No	One loop cannot carry 2 A and 5 A
(e) 10 V with 5 A	Yes	A current source accepts any voltage

OHM'S LAW

The sign of $v = iR$ depends on the reference arrow



(a) $v = iR$



(b) $v = -iR$

$$v = iR \qquad i = \frac{v}{R} \qquad G = \frac{1}{R} \text{ (S)}$$

The same rule as Chapter 1: check which terminal the arrow enters. Conductance G is measured in siemens (S).

POWER IN A RESISTOR

A resistor always absorbs power

$$p = i^2 R = \frac{v^2}{R} \geq 0$$

Whatever the arrows

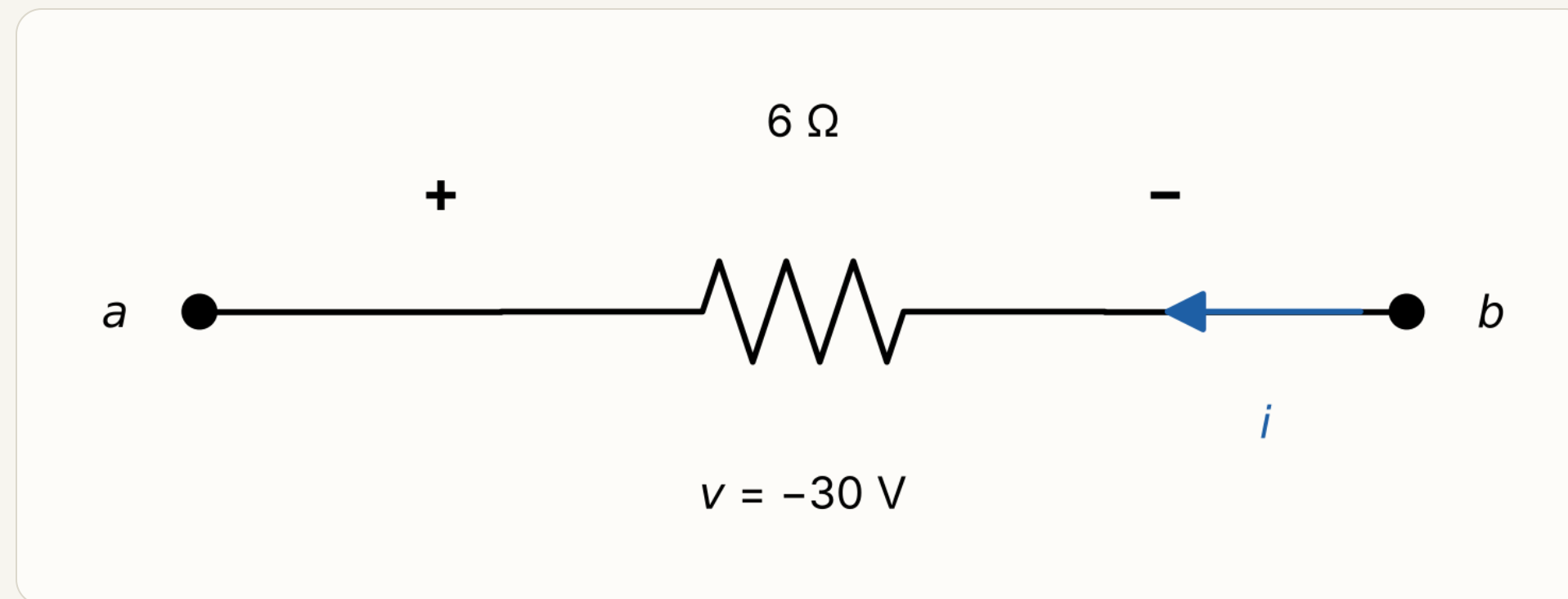
With the passive sign convention, $p = vi$, and for a resistor that is never negative: it turns electrical energy into heat.

A 2000 W, 220 V heater

$R = 24.2 \, \Omega$ and $i = 9.1 \, \text{A}$, so it trips a 5 A generator subscription on its own.

PREDICT FIRST • 1 MINUTE

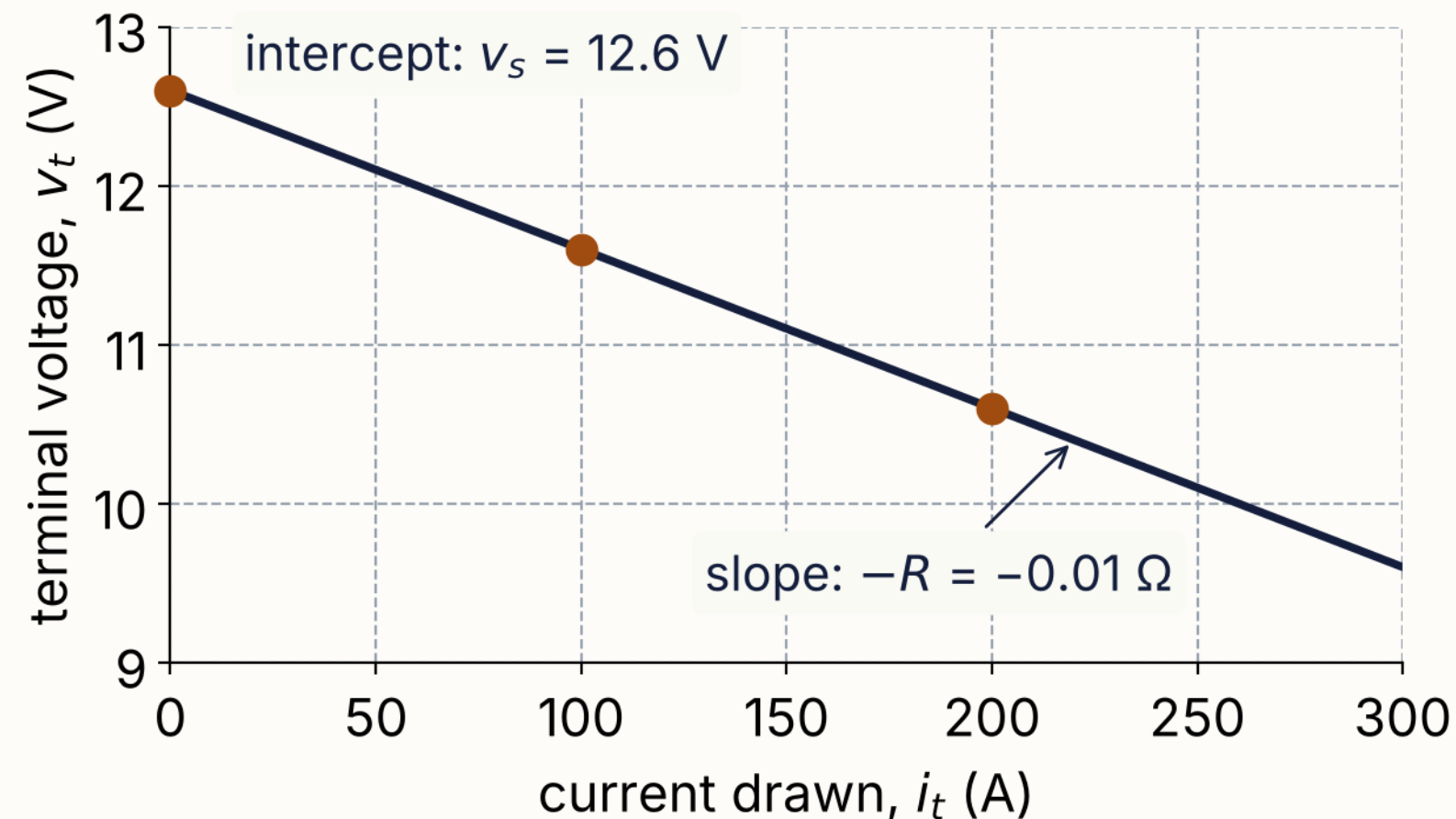
Find i . Does positive charge move from a to b, or from b to a?



$$v = -iR \Rightarrow i = -\frac{v}{R} = -\frac{-30}{6} = 5\text{ A}$$

The arrow enters the – terminal. $i = 5\text{ A}$, so charge moves along the arrow, from b to a : b is 30 V higher than a .

A real source: an ideal source plus a resistance



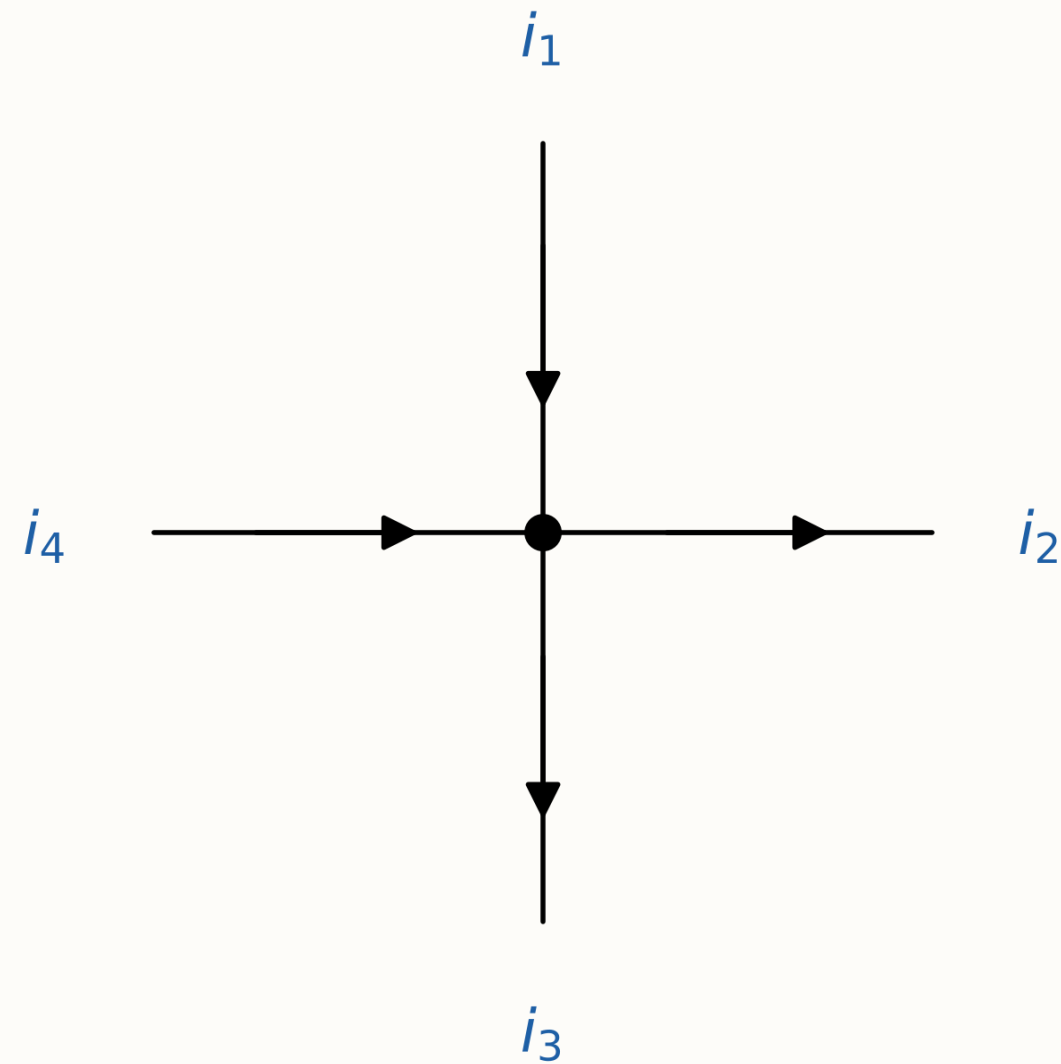
$$v_t = v_s - R i_t$$

Measure the terminal voltage at two currents: the intercept is the ideal source, the slope is the internal resistance.

Practice item S15 asks what the starter motor does to this battery.

KIRCHHOFF'S CURRENT LAW

KCL: the currents leaving a node add up to zero



$$\sum_{\text{leaving}} i = 0$$

$$-i_1 + i_2 + i_3 - i_4 = 0$$

A node is a point where two or more elements join. Count currents leaving as +, entering as -. Charge cannot pile up at a node.

KIRCHHOFF'S VOLTAGE LAW

KVL: the voltage drops around any closed path add up to zero

$$\sum_{\text{closed path}} v_{\text{drop}} = 0$$

How to write it

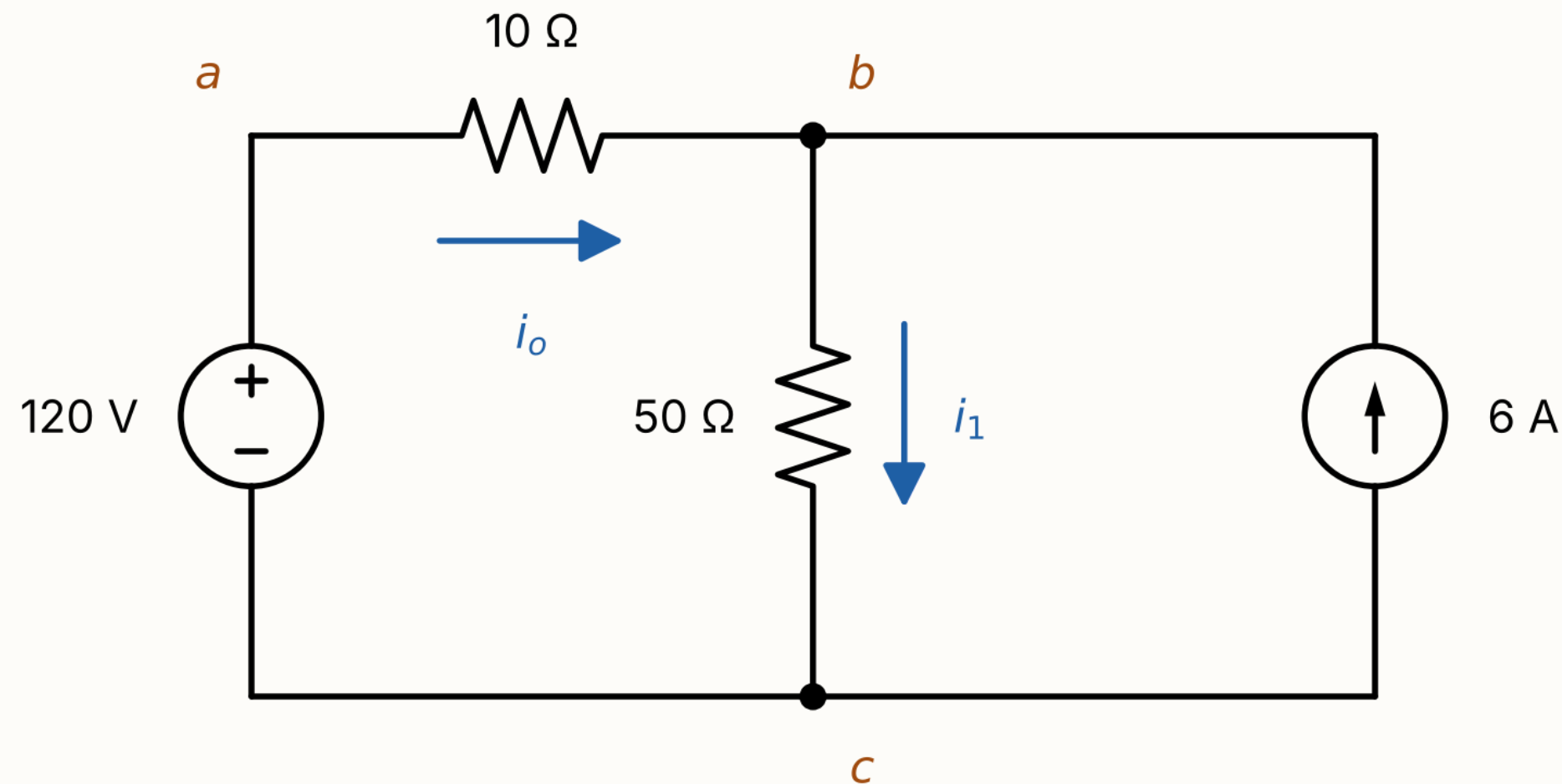
Walk the path in one direction. Crossing an element from + to – is a drop (+); from – to + is a rise (–).

Why it holds

A charge that goes round a closed path and returns to its start has gained and lost the same energy.

NILSSON EXAMPLE 2.8

Two unknown currents need two equations



Unknowns: the currents in the 10 Ω and the 50 Ω resistors.

KCL at b, currents leaving:

$$i_1 - i_o - 6 = 0$$

KVL clockwise around c-a-b-c, drops positive:

$$-120 + 10 i_o + 50 i_1 = 0$$

NILSSON EXAMPLE 2.8

Solve, then check with a power balance

$i_o = -3\text{ A}, \quad i_1 = 3\text{ A}$

Element	Power	Absorbs or delivers
10 Ω resistor	90 W	absorbs
50 Ω resistor	450 W	absorbs
120 V source	+360 W	absorbs
6 A source	−900 W	delivers

Delivered 900 W = absorbed 90 + 450 + 360 W. The negative current is not an error: the 120 V source is being charged.

STRATEGY

Choose your equations before you write them

Count

One equation for each unknown current or voltage.

KCL

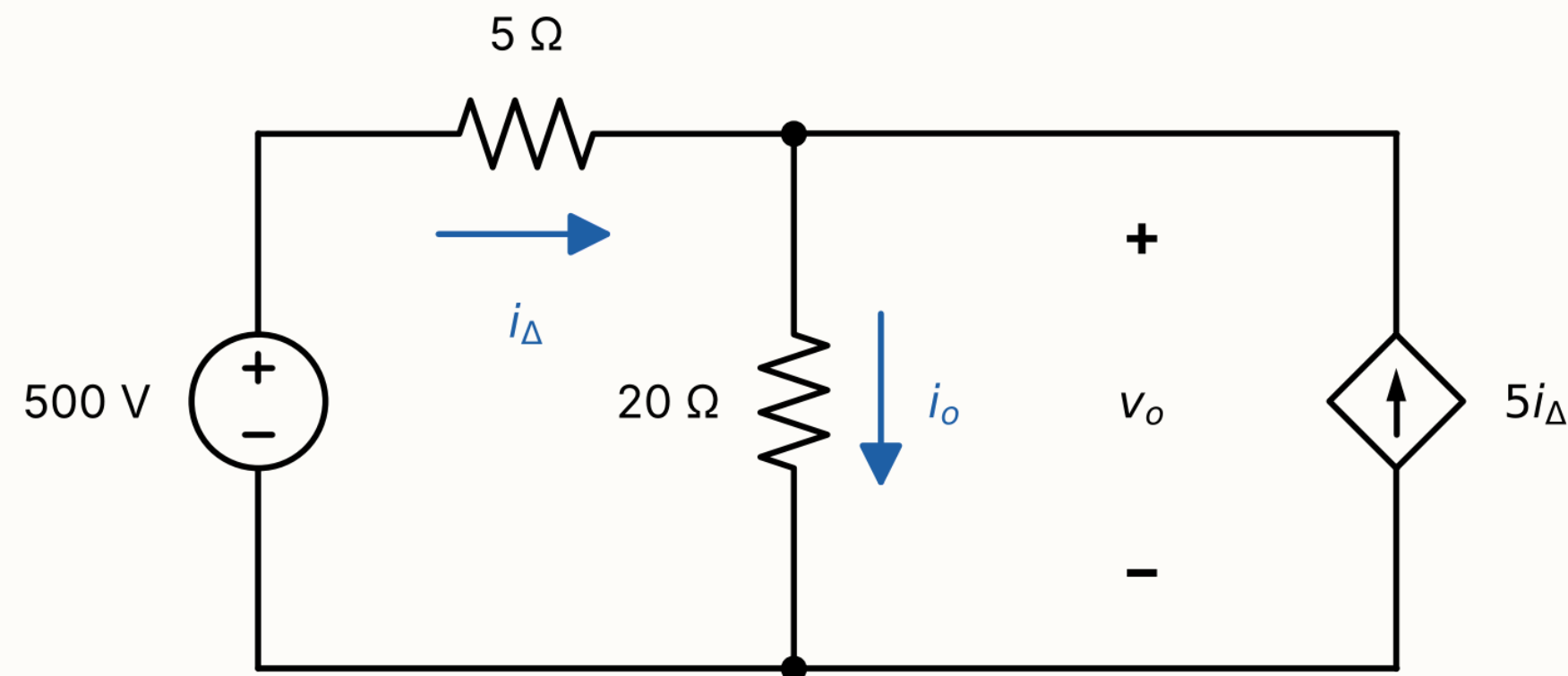
Use all nodes but one: the last node repeats the others.

KVL

Skip any loop through a current source: its voltage is one more unknown.

NILSSON EXAMPLE 2.10

A dependent source adds a constraint, not a new unknown



KVL around the left loop:

$$-500 + 5 i_{\Delta} + 20 i_o = 0$$

KCL at the top node, currents leaving:

$$-i_{\Delta} + i_o - 5 i_{\Delta} = 0 \Rightarrow i_o = 6 i_{\Delta}$$

The dependent source's current enters the node, so it counts with a minus sign.

NILSSON EXAMPLE 2.10

Substitute the constraint, then check the powers

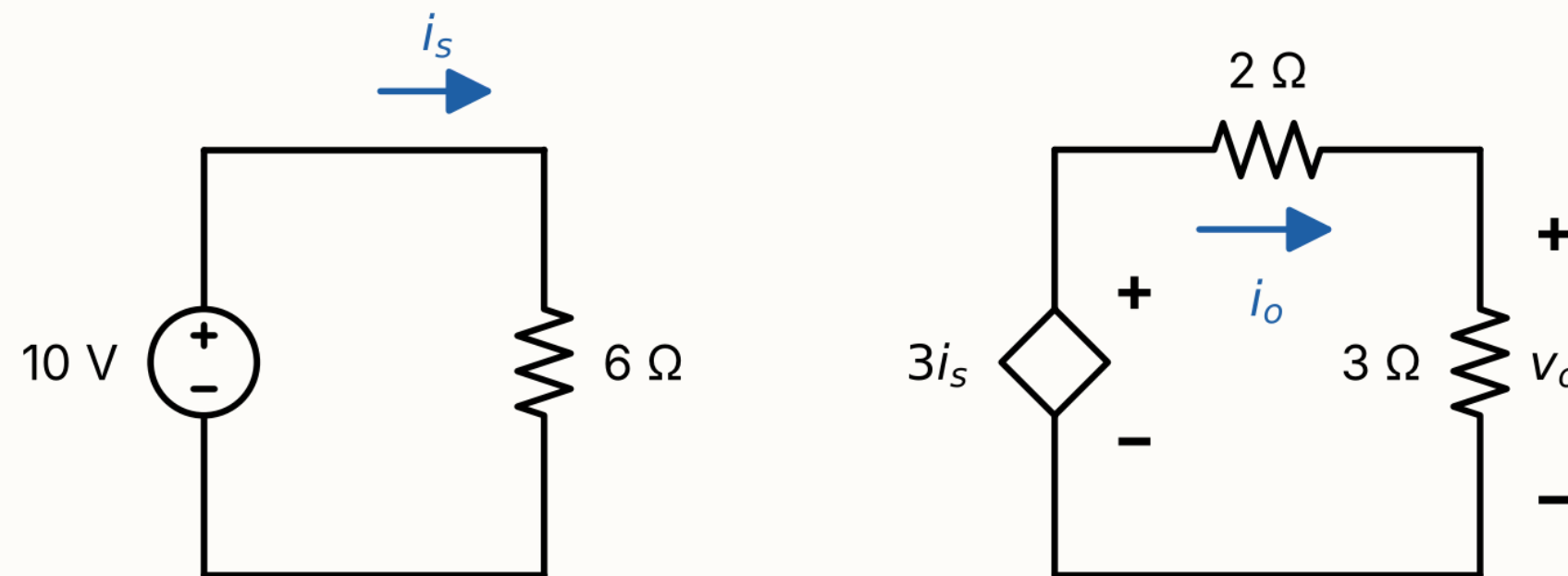
$$500 = 5 i_{\Delta} + 20 (6 i_{\Delta}) = 125 i_{\Delta}$$

$$i_{\Delta} = 4 \text{ A}, \quad i_o = 24 \text{ A}, \quad v_o = 20 i_o = 480 \text{ V}$$

Element	Power	Absorbs or delivers
500 V source	−2000 W	delivers
5 Ω resistor	80 W	absorbs
20 Ω resistor	11 520 W	absorbs
Dependent source	−9600 W	delivers

NILSSON EXAMPLE 2.11

A current-controlled voltage source



KVL, left loop:

$$-10 + 6 i_s = 0 \Rightarrow i_s = \frac{5}{3} \text{ A}$$

KVL, right loop:

$$-3 i_s + 2 i_o + 3 i_o = 0 \Rightarrow i_o = 1 \text{ A}$$

$$v_o = 3 i_o = 3 \text{ V}$$

The dependent source delivers 5 W.

Delivered: $16.7 + 5 = 21.7 \text{ W}$.

Absorbed: $16.7 + 2 + 3 = 21.7 \text{ W}$.

CHECKING AN AI ANSWER

An AI wrote this KCL equation for Example 2.10. What is wrong?

$$-i_{\Delta} + i_o + 5i_{\Delta} = 0$$

Correct:

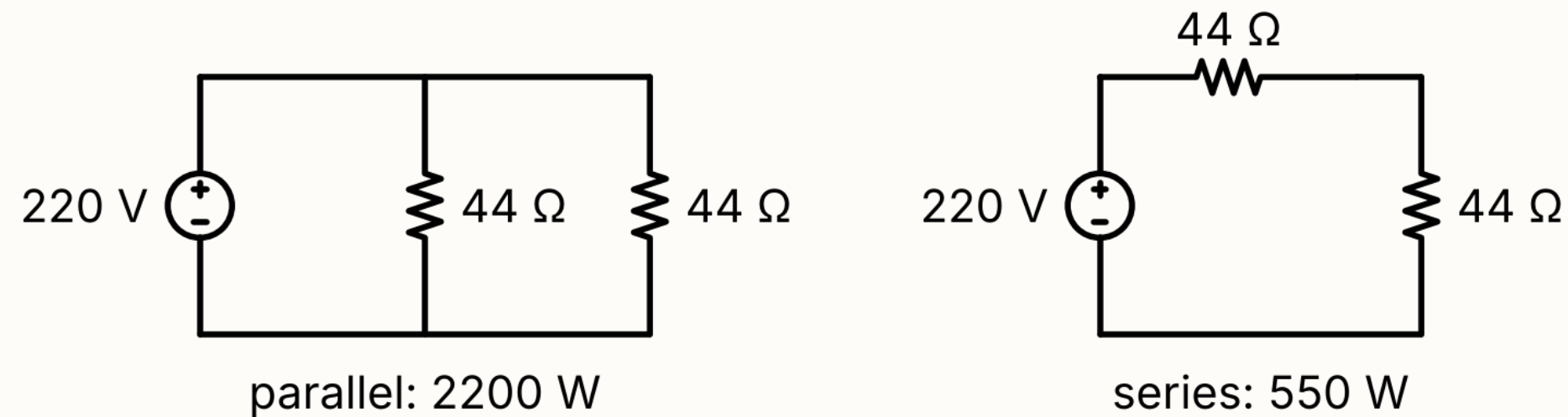
$$-i_{\Delta} + i_o - 5i_{\Delta} = 0$$

The dependent source's current **enters** the node, so it takes a minus sign. The giveaway: with the wrong sign the power balance fails.

In a 2025 study of Gemini 2.5 Pro on undergraduate circuit problems, misread source polarities caused about a third of its wrong answers, and misread current directions nearly as many.

BACK TO THE HEATERS

Series heaters give a quarter of the heat



$$R = \frac{v^2}{P} = \frac{220^2}{1100} = 44 \, \Omega \quad \text{series: } i = \frac{220}{88} = 2.5 \, \text{A}, \quad P = (220)(2.5) = 550 \, \text{W}$$

Parallel: each heater gets 220 V and gives 1100 W. Series: the resistance doubles and the current halves.

YOUR SELF-CHECK

$$\sum p = 0$$

Every source and every resistor, dependent sources included. If the powers do not add to zero, a sign is wrong.

Every booklet problem in this chapter ends with a power balance.

WHAT COMES NEXT

Practise openly, then show it on your own

Now, in class

Booklet Problems 2.1 to 2.7.
Problem 2.8 is homework.

At home

The 16-item practice sheet. AI
tools allowed; not marked.

Week 3

15-minute quiz, no devices.
Every question is a twin of a
practice-sheet question.

If you can do the sheet on your own, you will do well on the quiz.

CHECKLIST

You can do these without help

- Say whether two ideal sources may be connected together, and why
- Name the four dependent sources and the unit of each gain
- Write Ohm's law with the right sign from a diagram
- Write KCL at a node and KVL around a loop with consistent signs
- Solve a circuit with two unknowns, including a dependent source
- Check every answer with a power balance